

Multi-dimensional characteristics and governance strategies of rumors about the COVID-19 pandemic

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Abstract—This study was based on the rumor data of the National Joint Anti- Rumor Online Platform, data mining and textual analysis were used to analyze the contents of rumors from the spatial, temporal, and semantic dimensions. The results indicated that the number of rumors showed a relatively consistent trend with the development of the pandemic, spatial distributions of rumors were relatively uneven, exhibiting a concentration in large cities. In addition, rumor keywords in different spatiotemporal contexts were all related to the themes of the prevailing outbreak situation and the prevention and control measures, however, the obvious differences were seen as well. The number of rumors was positively correlated with the number of confirmed cases and was easily affected by external factors.

Keywords—COVID-19, Pandemic, Rumors, Spatiotemporal distribution, Analysis of characteristics, Visualization

I. INTRODUCTION

Rumors, which are generally considered to be fabricated statements with no factual basis[1], are an important research topic in the social and computer sciences[2]. The advent of major public events is often accompanied by the generation and propagation of rumors. Coronavirus Disease 2019 (COVID-19) developed into a global pandemic, accompanied by an infodemic. A large number of rumors have been propagated widely across social media platforms and throughout the internet. Their propagation brought panic and instability to society and seriously interfered with efforts to prevent and control the pandemic. Proper information dissemination has been found to be of critical importance in public health emergencies. The propagation of rumor at these times exhibits unusual characteristics that are substantially different from those appearing under normal social conditions[3][4]. Although the pandemic has entered a normalized stage in China, sporadic clusters of cases continue to emerge in various regions. This, combined with the interweaving of the factors of viral mutations,

vaccines, vaccinations, and imported cases, made the risk of rumor spread more complex and volatile. Thus, the analysis of rumors on the COVID-19 pandemic has social significance and important research value.

Researchers have conducted quantitative research on the propagation of rumors concerning the COVID-19 pandemic on the internet. Islam et al. compiled 311 pandemic-related rumors and reported conspiracy theories gathered from social media platforms such as Twitter and Facebook and then analyzed their contents[5]. Cinelli et al. collated pandemic-related rumors from six social media platforms outside China and conducted a comparative analysis of the modes of their propagation[6]. Chen et al. performed content analysis after gathering pandemic-related rumors from the Sina Weibo Official Account to Refute Rumors[7]. A comprehensive analysis of rumor propagation was performed by Chen et al. from the level of the basic elements, such as the propagated contents, propagators, recipients, and propagation effects[8]. Solovev et al. investigated whether COVID-19 misinformation spreads more viral than the truth and whether the differences in the diffusion of true vs. false rumors can be explained by the moral emotions they carry[9].

Using the results of previous research as a basis, this study investigated the contents of rumors from a spatiotemporal (ST) and semantic perspective. This was done to reveal the distributional and thematic characteristics of rumors in various ST contexts and discuss the relationship between the number of rumors in different ST contexts and the confirmed cases. At the same time, data on the rumors selected in this study spanned a long period of time, which led to a comprehensive understanding of the distributional ST characteristics of online rumors and popular topics during the pandemic. The results of this study could provide recommendations for the government

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and health authorities as they refine the process of rumor management.

II. RESEARCH DESIGN

A. Research framework

Octoparse web crawler was first used to obtain data on rumors from the National Joint Anti-Rumor Online Platform (NJAROP), pandemic data from the National Health Commission of China, and real-time pandemic big data from Baidu. Next, textual preprocessing was performed, which included data cleansing, segmentation of Chinese words, and filtering of stopwords. Subsequently, the distributional and semantic characteristics of the online rumors that were prevalent during the COVID-19 pandemic were examined in their spatial, temporal, and semantic dimensions, the results were visualized for display using infographics. Finally, strategies and suggestions for managing rumors are proposed. The overall research framework is shown in Figure 1.

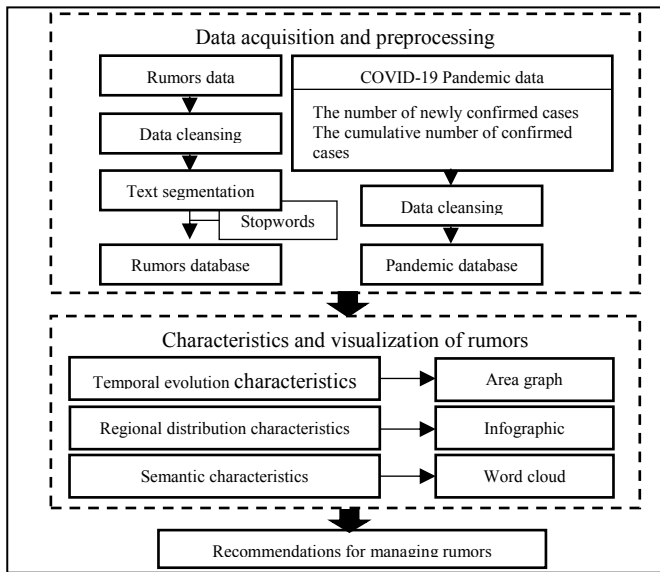


Fig. 1. Process research framework

B. Data acquisition and preprocessing

The research samples were data from the NJAORP's Prevention, Control, and Refutation Zone for COVID-19 pandemic rumors. Unlike other similar entities developed to refute rumors, the NJAORP's platform collates authoritative rumor-refuting information from relevant agencies, government departments at all levels, mainstream media, and experts. It provides detailed information on the contents of rumors and the timing of the publication and refutation of those rumors. Rumors carry a certain degree of representativeness and persuasiveness. Therefore, the rumor data selected for web crawling were those published between January 20, 2020 and May 31, 2022. After eliminating information that had been confirmed as fact and repeated information, a total of 1,081 rumor items remained, which served as the research sample.

C. Categories for analysis and coding

The rumor contents were divided into the four categories stated below for coding.

1) *Temporal category*: Some of the vaguely expressed data were verified based on the times at which the rumors circulated and were refuted. Vague time indicators, such as "the last few days" or "recently," or for which time data could not be verified, were eliminated. This was done to reflect the range of rumors at different stages of the pandemic, as well as the trends of changing themes.

2) *Regional category*: The geographic areas discussed in the rumors can be divided into provincial and city (county) levels, based on the divisions of China's administrative regions. This was done to reflect the range of rumors relating to various provinces and cities. The small amount of rumors involving multiple countries or provinces and cities were categorized in relation to their main subject.

3) *Thematic category*: The crawled data on rumors were combined with existing research results before the themes were divided into six categories, namely, (a) current state of the pandemic, (b) prevention and control measures, (c) virus information, (d) resumption of work and school, (e) international efforts to combat the virus, and (f) others. Two annotators were responsible for annotating the theme of the rumors based on the titles, and the two showed a 95% consistency. Ambiguous sentences were duly discussed until all the data were annotated to reflect the rumor themes at various ST contexts.

4) *Confirmed cases category*: This included the daily number of newly confirmed cases at various stages of the pandemic, as well as the cumulative number of confirmed cases in the areas referred to in the rumors. This was done to reflect the correlation between the prevalence of rumors and the confirmed cases.

III. ANALYSIS OF RESULTS

A. Temporal evolution of the number of rumors: Characteristics and visualization

The pandemic was divided into four stages according to the outbreak situation, using the white paper entitled Fighting COVID-19: China in Action[10], namely, outbreak (20 January to 20 February 2020), control (21 February to 17 March 2020), subsidence (18 March to 28 April 2020), and normalization (Since 29 April 2020). Trends in the number of rumors following changes to the state of the outbreak were analyzed according to these stages (Figure 2). The specific details of the analysis are given below.

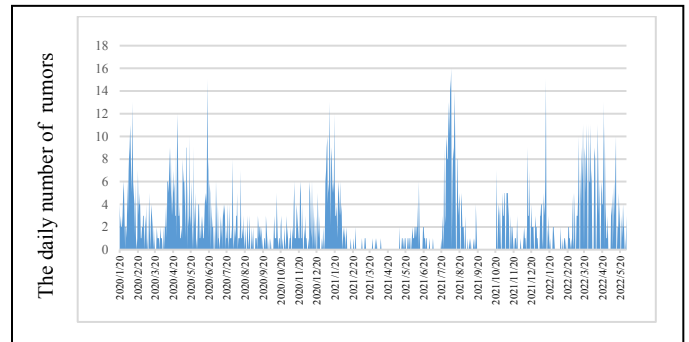


Fig. 2. Trends in the number of rumors at different stages of the pandemic.

The overall trend of the changing number of rumors indicated concentrations in the periods of February and April 2020, when the pandemic was in the outbreak and subsidence stages. From the beginning of May, the numbers then began to follow an overall downward trend. However, there were short-term rebounds and fluctuations due to small-scale outbreaks in localized areas. During the research period, the highest values were reached in June 2020, January and early August 2021, and March to April 2022. The possible factors behind these changes are specifically analyzed while the pandemic situation is being considered, and they are elaborated upon below.

In the initial stages of the outbreak, most of the general public was oblivious to the unfolding situation. A small amount of rumor-refuting information on the pandemic began appearing on the NJAORP beginning in January 20, 2020. With the number of newly confirmed cases rising nationwide, rumors also increased slightly, starting at the beginning of February. When the cumulative number of confirmed cases reported across the country reached 15,152 as of February 12, the number of rumors reached its peak value within the sampling period. After the pandemic was gradually brought under control in Hubei and Wuhan, as of February 21, panic among the general public began to subside. The number of rumors also significantly declined, thanks to the reduced danger.

During the subsidence stage, the number of daily new cases in China was kept to the single digits, and the combat against the virus reached a phased success. However, the number of rumors did not decline, in spite of the effective suppression of the pandemic. Instead, a significant increase appeared from April 12 to 28. The reasons had two aspects. First, the situation in northern China quickly rebounded, and the pandemic had been on-going for a long time, leading to a growth rise in the number of rumors in this region. Second, the pandemic was at the outbreak stage in other countries, leading to a rise in rumors related to international prevention and control.

In China, at the beginning of May, the pandemic situation was sporadic and scattered in general. There were clusters of cases appearing in some areas that were triggered by sporadic cases, whereas imported cases were basically under control. By this point, pandemic prevention and control had entered the normalization stage, and the number of rumors presented an overall downward trend as the situation continued to improve. Nevertheless, fluctuations were seen, due to rebounding numbers of cases in localized areas. Epidemic data show that there are four clearly volatile period in normalization stage.

- June 2020, a cluster of cases occurring at Beijing's Xinfadi Wholesale Market, together with sporadic related cases appearing in Hebei and Liaoning. This led to a corresponding rise in the number of rumors circulated in these regions.
- January 2021 was also a unique temporal node: multiple factors coming together led to the breeding of additional pandemic-related rumors. A central factor was the serious situation in central and northern China and Sichuan from mid-December 2020 to January 2021, resulting in Shijiazhuang in Hebei, Suihua in Heilongjiang, and other places being listed as high-risk.

The number of medium-risk areas also rose to 60. At the same time, the number of newly confirmed patients in these areas grew sharply. Another factor here was the approaching Lunar New Year celebrations. Thanks to the large flows of migrant workers returning to their hometowns, rumors related to the "holidays" and "being quarantined" increased. There were also general concerns regarding the vaccines being used in the vaccination programs, indicated in rumors related to "vaccines" and whether they could prevent viral infections. The number of rumors again increased when the first batch of COVID-19 vaccines were successfully researched, developed, and then administered to the population.

- During July and August 2021, the outbreak was caused by the imported Delta variant from Nanjing Lukou International Airport, virus flares in more than 20 cities across the country, especially as rumors increased in areas with serious outbreaks such as Nanjing and Yangzhou.
- Between March and May 2022, the highly infectious Omicron variant spread to 29 provinces, with serious outbreaks in Shanghai, Jilin, and Hangzhou, and rumors during this period increased significantly and lasted longer.

In order to examine the relationship between the number of rumors and the rate of newly confirmed cases, SPSS software was used to perform a Pearson's correlation analysis between two sets of numbers (statistics every 7 days). These showed a positive correlation in their quantitative characteristics ($r = 0.424$, $P < 0.001$), indicating that the number of rumors was relatively consistent with the trend of the pandemic development. However, the rumors were also easily affected by external factors.

B. Regional distribution of rumors: Characteristics and visualization

The number of rumors in the various provinces and cities was analyzed to obtain a comprehensive understanding of their distribution at different spatial scales. The correlation between the number of rumors at a spatial scale and the number of confirmed cases was also analyzed. There were 1449 rumors involving 31 provinces (or municipalities or autonomous regions). All of China's hinterland provinces were included, showing that the pandemic received widespread attention across the country. It can be seen from the histogram in Figure 3 that the numbers of rumors in Shanghai, Zhejiang, Jiangsu were significantly higher than those in other regions. The numbers in Hebei, Hubei, Shandong, Beijing, Sichuan, Guangdong, Hunan, and Henan were also relatively high. Qinghai, Guizhou, Ningxia, and Xizang had low numbers. The numbers in the other 16 provinces were at moderate or relatively low levels. The number of rumors regarding the top 11 provinces accounted for 71% of the total number, indicating that the spatial distribution was relatively uneven. Coincidentally, most of those 11 provinces are the site of the country's densely populated and economically developed areas.

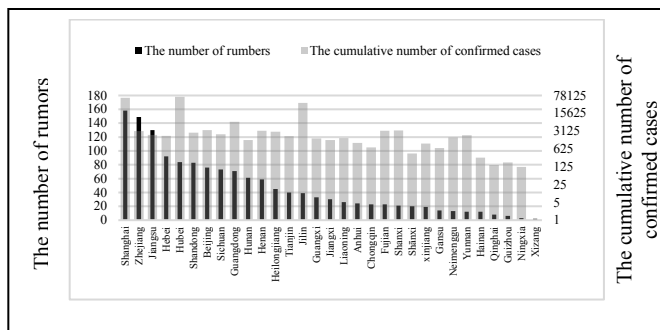


Fig. 3. The number of rumors and cumulative confirmed cases in the various provinces.

Using the analytical results as a basis, SPSS software was used again to analyze the Pearson's correlation between the number of rumors in each province and the cumulative number of confirmed cases as of May 31, 2022 ($r = 0.394$, $P < 0.05$). Figure 2 also shows that the rumors and cumulative confirmed cases do not correspond for some provinces. This indicates a weak correlation between the number of rumors in a province and its cumulative number of confirmed cases. These contradictions were more apparent for the Hebei and Jilin regions.

From the analysis of distribution of rumors across cities, there were 329 cities showing 899 rumors in total. Among them, 179 prefecture-level cities featured 711 rumors, accounting for 79% of the total rumors at the city level; the 150 county-level cities featured 188 rumors, accounting for approximately 21% of the total number. Preliminary indications showed that rumors were mainly propagated in the prefecture-level cities. According to the statistical data, it can clearly be seen that the number for Nanjing and Wuhan was significantly higher than that of the other cities. It was followed by Chengdu, Shenzhen, Hangzhou, and Shijiazhuang. There were 267 cities with a low number of rumors (two or less). Thus, there were clear disparities existed in the distribution of the number of rumors across different cities, with a coefficient of variation larger than 1.

The economic and population data further showed that the top ten cities in terms of number of rumors were mostly provincial capitals or sub-central cities, which are economically more developed and more densely populated. A pattern of likely concentrations was seen in the large cities. An exception to the case was Ruili, which showed the largest number of rumors among the county-level cities in the country. A possible reason for this is that it is located near the Chinese–Myanmar border where nine outbreaks had occurred in the past two years, complicating the prevention and control situation.

To further determine the relationship between the number of rumors in different cities and the outbreak situation, Pearson's correlation analysis was performed between the number of rumors in a city and its cumulative number of confirmed cases ($r = 0.651$, $P < 0.001$). The results indicated a significant correlation between the two sets of numbers. In other words, cities that featured many related rumors usually had more severe outbreaks.

In summary, the distribution of the number of rumors at different spatial scales was relatively uneven, reflecting the phenomenon that rumors were more likely to be propagated in large cities that are more densely populated and more developed economically. Although provinces and cities that featured more rumors tended to also be areas with more severe outbreaks, there were some abnormally distributed values. This was a reminder that in addition to the pandemic, other factors—including population, the economy, education, and prevention and control measures for the situation—could also affect the propagation of rumors in cities.

C. Semantic characteristics of the rumors: Analysis and visualization

The software program ROST was used in this study to process word segmentations and count the word frequencies in the title of rumors. The top 20 most frequent words were selected to prepare a word cloud diagram, the visualization tool for creating word clouds is EdWordle. High-frequency keywords were then applied to explore the differences and connections between the semantic features of rumors at different stages of the development of the pandemic and in various provinces.

1) *Semantics at each stage of the pandemic*: The statistical results for word frequencies indicated that keywords at all stages of the pandemic exhibited two main characteristics (Figure 4). First, high-frequency keywords—"novel coronavirus pneumonia," "confirmed diagnosis," and "patient cases"—were associated with various stages of the development of the pandemic, reflecting that the public was constantly concerned with the prevailing outbreak situation. Second, the keywords presented a phased distribution mirroring the developing situation, indicating that rumors with varying themes were prevalent at different stages. The typical differences were reflected in several aspects.

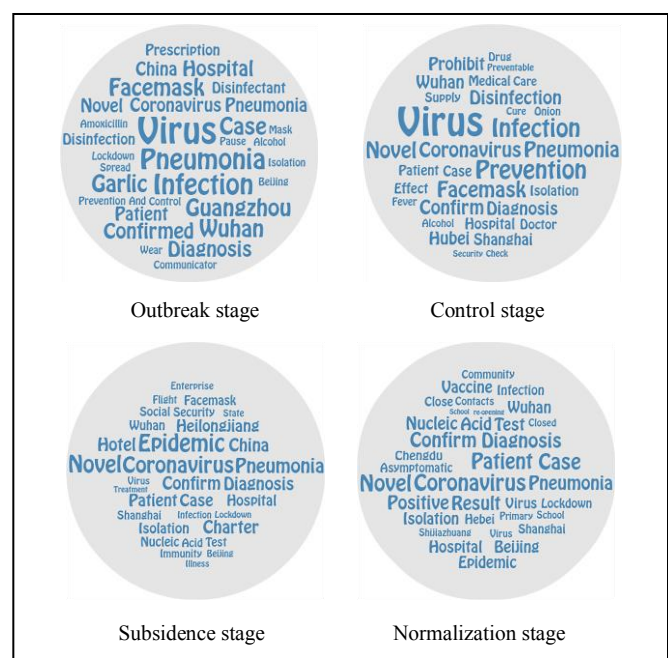


Fig. 4. High-frequency keywords at different stages.

During the outbreak of the pandemic, the highest-frequency keywords included “prevention,” “masks,” and “disinfection.” Rumors at this stage focused on prevention and control measures for, and methods to protect against, COVID-19. At the control stage, the high-frequency words were “prevention,” “Beijing,” “transmission,” and “Wuhan,” indicating that in addition to the earlier themes was that of viral transmission. Examples of these included “COVID-19 is transmitted vertically from mother to baby” and “COVID-19 can be transmitted through mosquito bites.” These reflected speculations about the ways through which the virus was transmitted. Besides, Beijing is the political, economic, and cultural center of China, and it has registered COVID-19 cases from the beginning of the outbreak of the pandemic. This led to a corresponding increase in rumors. Some residents of Wuhan experienced psychological trauma caused by the severity of the outbreak there, and rumors related to mental health increased correspondingly. After the pandemic subsided, the high-frequency words included “China,” “being quarantined,” “school re-opening,” and “foreigners.” As outbreaks occurred in other countries, a corresponding increase in rumors was seen, centered on international hospitals combating the virus, imported cases from overseas, and foreign personnel being quarantined. As control of the pandemic in China became more effective, speculations about work resumption and school re-opening increased correspondingly.

As the normalization phase of the pandemic became prolonged, the keywords for rumors showed the characteristics of phased changes. First, those related to “positive results” and the “nucleic acid test” increased from the beginning of May 2020, which could be related to the country’s promotion of nucleic acid tests as a prevention and control measure. Second, “Beijing,” “Hebei,” “Shanghai,” “Zhejiang,” “Nanjing,” and other place names indicating a clear subject appeared among the high-frequency words. This was related to the situation in the country where an interweaving of sporadic and scattered cases was seen together with localized clusters. These provinces (cities) also showed relatively severe outbreaks. In addition, “COVID-19 vaccines” became a high-frequency word in August 2020, in tandem with rising speculations about the effectiveness of the vaccines, their cost, and when they would be available. Finally, keywords such as “asymptomatic,” “infected persons,” and “close contacts” appeared frequently, reflecting that asymptomatic infected persons and close contacts were risk points that could trigger outbreaks.

2) *Semantics of the typical regions:* Rumors from 11 typical provinces were selected for word frequency analysis. The selection was based on the distribution of the rumors at various spatial scales. The data reflected similarities and differences between keywords associated with the various provinces, as analyzed below.

Three main types of similar words were identified after a comparison of the high-frequency words from the various provinces. First, the most frequent word in all the provinces was “COVID-19,” closely followed by words such as “confirmed diagnosis,” “patient cases,” “nucleic acid test,” and “positive results.” These reflected the public’s concerns about the prevailing situation of the pandemic. Second, the keyword “hospital” was found to be distributed across all provinces. Textual co-occurrence analysis showed that the word “hospital”

was generally associated with “COVID-19,” “patients,” “close contacts,” and “medical personnel.” A possible reason was the belief that people could be easily infected by the virus in a hospital environment. Third, related keywords such as “lockdown,” “prohibit,” and “cease operations” accounted for a definite proportion, attracting public attention to the prevention and control measures during the outbreaks. In addition, the place names of cities related to the pandemic appeared in rumors in different provinces. “Wuhan,” “Shanghai,” being the first epicenter of the pandemic, was the city name that had the highest frequency. Other words with relatively high frequencies included “Nanjing,” “Chengdu,” “Guangzhou,” and “Shenzhen.” These also had significantly higher frequencies than other cities in the province to which they belong.

Variations appeared in the distribution of the keywords of rumors in different provinces. First, among the keywords for the two provinces (cities) of Beijing and Jiangsu, “school,” “school re-opening,” and “students” had the highest frequency. This reflected that people in these areas were speculating about outbreaks in schools, re-opening of schools for students, and when examinations would be held. Second, words related to “frozen” and “food” were more frequent in rumors of the three provinces (cities) of Shandong, Beijing, and Heilongjiang. This might be because the outbreaks in these places were all related to imported cold-chain food. Third, in the rumors in Shanghai, the words “supplies,” “makeshift hospital,” “confirmed case” and “lockdown” were frequently used, because Shanghai was faced with the worst covid-19 outbreak in the past two years. Fourth, because it is the capital of China, Beijing featured most cross-regional rumors. It was not only associated with other Chinese cities such as “Tianjin,” “Hebei,” and “Shanghai” but also to international rumors related to the combating of the virus.

IV. CONCLUSION AND RECOMMENDATIONS

A. Research conclusion

In this study, data on rumors were obtained from the NJAORP, and data mining and textual analysis were used to determine the distribution and semantic characteristics of the number of rumors in various ST dimensions, as well as the relationship between the number of rumors in different ST contexts and confirmed cases. Three meaningful conclusions were drawn from the research results and are stated below.

First, after the characteristics of the temporal distribution of the number of rumors were analyzed, a relatively consistent trend was identified between the numbers and the general development of and changes in the pandemic. However, the numbers were easily affected by external factors, and unusual fluctuations in the numbers could still appear even when the outbreaks were well controlled. Factors that were likely to trigger an increase in the number of rumors included small-scale outbreaks, changes in the international state of the pandemic, and prevention and control measures.

Second, the spatial distribution of the rumors was seen to extend across 329 cities in 31 of China’s provinces (as well as municipalities and autonomous regions). Despite the vast regional coverage, the numbers were unevenly distributed at the provincial and municipal scales. This reflected the phenomenon that rumors were more likely to propagate in the large cities,

which are more densely populated and more developed economically. In addition, the number of rumors at various spatial scales was positively correlated with confirmed cases, indicating that the numbers were related to the severity of the regional outbreaks.

Third, the analysis of the semantic characteristics of the rumors showed that the keywords of rumors in different ST contexts were all related to the themes of the prevailing outbreak situation along with the prevention and control measures. However, variations were seen in the keywords of rumors under different ST contexts. The analysis of the temporal evolution of the keywords indicated that they exhibited phased characteristics that were associated with changes in the outbreak situation. This reflects the general public's concerns over related themes such as prevention and protection against the virus, information on the virus, and prevention of outbreaks at the international level. It was noted that the names of cities with severe outbreaks appeared as high-frequency keywords. This reflected concerns over the themes of viral transmission and the re-opening of school for students.

B. Recommendations for managing rumors

Following the research conclusions indicated above, three recommendations are proposed to manage rumors. First, social concerns vary at different stages of a pandemic or other event, so government departments and mainstream media must publish accurate and detailed information on outbreaks to inform the general public in a timely and dynamic manner. The general public's social concerns should receive expedited attention. Dismantling the public's belief in rumors requires the government and mainstream media to establish a reliable and authoritative institutional image. A mechanism for well-founded trust between the general public, government, and mainstream media should be established to effectively curb the propagation of rumors.

Second, given the uneven distribution of the number of rumors in different provinces or cities, relevant governing bodies departments must strengthen the monitoring and analysis of rumors in key outbreak areas, especially in the more densely populated and more economically developed large cities. This will improve the level of refined analysis of the spatial distributional characteristics of rumors, which will facilitate the formulation of differentiated and precise strategies to dispel rumors in the key areas. This, in turn, will lead to the effective management of rumors and prevention of their propagation.

Third, scientific knowledge on outbreak prevention should be vigorously disseminated to enhance the general public's scientific literacy is enhanced, and they are better equipped to combat the virus. As COVID-19 continues to mutate, the pandemic may need to be normalized. The public should be guided to master basic knowledge on health and disease prevention, establish awareness for self-protection, and respond rationally to future changes in the pandemic.

In summary, data mining and textual analysis were used in this study to examine the contents of rumors from the ST perspective, thereby revealing the distributional and semantic characteristics of rumors across different ST dimensions. Relevant strategies for improved rumor management and prevention were then proposed based on the research conclusions. To the best of our knowledge, this study first quantitatively analyzes COVID-19 related rumors from the different ST dimensions. It provides a more comprehensive and profound understanding of rumor propagation in different stages and key areas and is of great value for rumor management in public emergencies. In follow-up studies, rumor themes will be classified in further detail, and the characteristics of the temporal evolution of rumors in different regions will be analyzed. Regional factors, such as population figures, per capita GDP, rate of internet penetration, and other variables will also be introduced to examine the circulation of rumors in cities, along with the relationship between those factors. This will lead to greater in-depth revelation of the characteristics of temporal changes of rumors in the key areas and the factors affecting those characteristics, thereby compensating for the inadequacies of this study.

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